

GANGHUA WANG

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Research Interests: Trustworthy AI, Deep learning theory and applications, Machine learning theory, Model fairness

PROFESSIONAL EXPERIENCE

University of Chicago, Data Science Institute
Postdoctoral Researcher

Sept. 2024 - present
Chicago, IL

EDUCATION

University of Minnesota, Twin Cities
Ph.D. in Statistics

Aug. 2019 - Aug. 2024
Minneapolis, MN

Advised by [Prof. Jie Ding](#) and co-advised by [Prof. Yuhong Yang](#)

Peking University
B.S. in Statistics, Minor in Economics

Sept. 2015 - July 2019
Beijing, China

PUBLICATIONS

Published

* indicates equal contributions

- [1] **Ganghua Wang**, Ali Payani, Myungjin Lee, and Ramana Kompella. “Federated learning with group bias mitigation: beyond the local fairness”. In: *Trans. Mach. Learn. Res.* (2024). [\[pdf\]](#).
- [2] **Ganghua Wang***, Xun Xian*, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “Demystifying Poisoning Backdoor Attacks from a Statistical Perspective”. In: *Proc. ICLR* (2024). [\[pdf\]](#).
- [3] Enmao Diao*, **Ganghua Wang***, Jie Ding, Yuhong Yang, and Vahid Tarokh. “Pruning deep neural networks from a sparsity perspective”. In: *Proc. ICLR* (2023). [\[pdf\]](#).
- [4] Gen Li, **Ganghua Wang**, and Jie Ding. “Provable Identifiability of Two-Layer ReLU Neural Networks via LASSO Regularization”. In: *IEEE Trans. Inf. Theory* 69.9 (2023), pp. 5921–5935. DOI: [10.1109/TIT.2023.3274152](#).
- [5] **Ganghua Wang**, Jie Ding, and Yuhong Yang. “Regression with Set-Valued Categorical Predictors”. In: *Statistica Sinica* 33.4 (2023), pp. 2545–2560. DOI: [10.5705/ss.202021.0332](#).
- [6] Xun Xian, **Ganghua Wang**, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “A Unified Framework for Inference-Stage Backdoor Defenses”. In: *Proc. NeurIPS* (2023). [\[pdf\]](#).
- [7] Xun Xian*, **Ganghua Wang***, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “Understanding backdoor attacks through the adaptability hypothesis”. In: *Proc. ICML* (2023). [\[pdf\]](#).

Under review

- [8] Xun Xian, **Ganghua Wang**, Xuan Bi, Jayanth Srinivasa, Ashish Kundu, Charles Fleming, Mingyi Hong, and Jie Ding. “On the Vulnerability of Applying Retrieval-Augmented Generation within Knowledge-Intensive Application Domains”. In: *Proc. ICLR* (2025).

- [9] Xun Xian, **Ganghua Wang**, Xuan Bi, Jayanth Srinivasa, Ashish Kundu, Mingyi Hong, and Jie Ding. “RAW: A Robust and Agile Plug-and-Play Watermark Framework with Provable Guarantees”. In: *Proc. ICLR* (2025).
- [10] **Ganghua Wang** and Jie Ding. “Subset Privacy: Draw from an Obfuscated Urn”. In: *arXiv preprint* (2024). [\[pdf\]](#).
- [11] Wenjing Yang*, **Ganghua Wang***, Jie Ding, and Yuhong Yang. “A Theoretical Understanding of Neural Network Compression from Sparse Linear Approximation”. In: *IEEE Trans. Pattern Anal. Mach. Intell., Under Major Revision* (2024). [\[pdf\]](#).

Manuscript

- [12] **Ganghua Wang**, Zhiyuan Tang, and Jie Ding. “Classification with set-valued labels”. In: *Preparation* (2024).
- [13] **Ganghua Wang**, Yuhong Yang, and Jie Ding. “Model Privacy: A Framework to Understand Model Stealing Attack and Defense”. In: *Preparation* (2024).

HONORS AND AWARDS

Awarded by University of Minnesota, Twin Cities

Doctoral Dissertation Fellowship Finalist	2023
School of Statistics Travel Award	2023
Summer Research Fellowship	2020
School of Statistics First Year Scholarship	2019

Awarded by Peking University

The Academic Excellence Scholarship (3 times)	2016 - 2018
Fang Zheng Scholarship	2017
Wu Si Scholarship	2016, 2018
Freshman Scholarship	2015

Awarded by Cisco Systems, Inc.

Cisco Research Graduate Award	2022
Cisco Research Fellow	2022, 2023

TEACHING

Teaching Assistant, University of Minnesota, Twin Cities

<i>STAT 4102 Theory of Statistics II</i>	Fall 2020
<i>STAT 3021 Introduction to Probability and Statistics</i>	Spring 2020
<i>STAT 3011 Introduction to Statistical Analysis</i>	Fall 2019

SERVICES

Member of

Graduate Student Liaison Committee, School of Statistics, University of Minnesota	June 2023 - Aug. 2024
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Mentor of

Undergraduate Reading Group, School of Statistics, University of Minnesota	2024
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AEOP High School Apprenticeship K-12 Outreach Program to enhance diversity, equity, and inclusion (sponsored by Army Research Office)	2022
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Reviewer of

AISTATS, AAAI, ICML, ICLR, ICASSP, Neural Processing Letter

SOFTWARE

Python Packages

- 1. **SubsetPrivacy**: Implement the subset privacy mechanisms proposed in [10] and statistical inference methods for set-valued observations [5].
- 2. **ModelPrivacy**: Implement the common and proposed model stealing defense/attack strategies on benchmark datasets [13].
- 3. **FedGFT**: Implement a bias mitigation federated learning algorithm proposed in [1]. It has been incorporated into **Flame**, an open-source project for federated learning systems, which is developed by Cisco Systems, Inc.
- 4. **CBD**: Conformal backdoor defense that detects the backdoor inputs in the inference stage with provable guarantees on false positive rate [6].
- 5. **RAW**: A robust and agile watermarking technique for AI-generated images [9].

PATENTS

- 1. G. Wang, M. Lee, A. Payani, R. Kompella, “Group Bias Mitigation in Federated Learning Systems,” 07/28/2023, US patent #18/227,535

TECHNICAL STRENGTHS

Computer Languages Pyth on, MATLAB, R, SQL

PRESENTATIONS AND TALKS

Clemson University, Clemson, SC	Sept. 2024
International Conference on Econometrics and Statistics, Beijing, China	July 2024
International Conference on Learning Representations, Vienna, Austria	May 2024
Department of Statistics, Chinese University of Hong Kong, Hong Kong	Apr. 2024
Conference on Neural Information Processing Systems, New Orleans, LA	Dec. 2023
School of Statistics, University of Minnesota, Minneapolis, MN	Nov. 2023
Joint Statistical Meeting, Toronto, Canada	Aug. 2023
International Conference on Econometrics and Statistics, Tokyo, Japan	Aug. 2023
International Conference on Machine Learning, Honolulu, HI	July 2023
Joint Conference on Statistics and Data Science, Beijing, China	July 2023
Center of Statistical Research,	June 2023
Southwestern University of Finance and Economics, Chengdu, China	
Center for Statistical Science, Peking University, Beijing, China	June 2023
International Conference on Learning Representations, Kigali, Rwanda	May 2023
School of Statistics Student Seminar, Minneapolis, MN	Mar. 2023